

VISHWAJIT MALL

Python Backend Engineer | AI / LLM Engineer | RAG & Agentic Systems

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SUMMARY

Python backend engineer with 4+ years of experience designing and shipping production APIs, with a growing specialization in applied AI systems: Retrieval-Augmented Generation, multi-provider LLM orchestration, and agentic tool-use architectures. Built and deployed multi-tenant FastAPI backends with async SQLAlchemy, JWT/OIDC authentication, Redis-backed caching and rate limiting, Celery background workers, and Docker-based multi-service deployments. Designed and published an open-source Python logging library on PyPI, now used across multiple production backends. Comfortable owning a system end-to-end — architecture, data modeling, API design, and observability.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, PHP, SQL

AI / LLM Engineering: LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Multi-Provider LLM Routing & Fallback (Gemini, Groq, OpenAI, Hugging Face, Ollama), Prompt Engineering, AI Agents (ReAct-style tool use), MCP-style Tool Risk Classification

Backend Development: FastAPI, Async Python, Node.js, Express.js, Laravel (PHP), REST API Design, GraphQL, Celery Background Workers

Databases & Vector Stores: PostgreSQL, MongoDB, MySQL, Qdrant, pgvector, Redis

Authentication & Security: JWT, OAuth / OIDC (Keycloak), Role-Based Access Control (RBAC)

DevOps & Infrastructure: Docker, Docker Compose, Nginx, GitHub Actions (CI/CD)

Observability: Structured Logging, Request Tracing, Prometheus, Grafana, Loki

Testing: pytest

Cloud: AWS S3, AWS RDS, AWS Amplify

Tools & Project Management: Git, GitHub, JIRA, Azure DevOps (Boards, Sprint & Work Item Tracking)

EXPERIENCE

Full Stack Developer — Bee Logical Software Solutions

Pune, India · Nov 2022 – Jun 2026

- Served as a core developer delivering production platforms end-to-end, including SmartHR, an AI-powered candidate interview automation system (technical details under Projects), alongside SetBot (property booking platform) and Bella Canvas (3D product customization & e-commerce).
- Received the company's Outstanding Performance Award (November 2024) for engineering contributions and team collaboration.

Software Developer — WebReinvent

New Delhi, India · Jul 2022 – Oct 2022

Full Stack Developer — BluBrandz Technologies Pvt Ltd

New Delhi, India · Aug 2021 – Jun 2022

- Built a hospital management system (Angle Fertility) to digitize patient records, appointment booking, and clinical workflows for healthcare staff.
- Developed an event management system (Aragma — House Party Management) for planning events and tracking participants, expenditures, and staff records.
- Built an equipment management system (Unicorn Equipments) with a Laravel-based admin dashboard for tracking and maintaining listed machinery.

PROJECTS

TenantMind-AI — Multi-Tenant AI Property Management Platform

An AI platform that lets property-management teams query lease documents in natural language and automate tenant workflows, with each organization's data kept isolated from the others.

FastAPI · Keycloak (OIDC) · MongoDB · Qdrant · Celery · Redis · Docker

Independent Project · github.com/vishwajitvm/TenantMind-AI

- Architected a multi-tenant FastAPI backend with Keycloak-based OpenID Connect authentication and per-tenant request context resolution.

- Built a multi-provider LLM gateway with automatic fallback across Gemini, Groq, OpenRouter, and Ollama, logging latency and token usage per call.
- Implemented a Retrieval-Augmented Generation endpoint using per-tenant Qdrant vector collections for lease-document grounding.

AgentHive — Multi-Agent Orchestration Platform

A platform for building AI agents that reason step by step, decide which tool to use, and carry out multi-step tasks on their own.

[FastAPI](#) · [PostgreSQL](#) / [pgvector](#) · [Celery](#) · [Redis](#) · [Docker](#)

Independent Project · github.com/vishwajitvm/AgentHive

- Built a ReAct-style agent execution loop supporting step-by-step reasoning, tool invocation, and full run/step/tool-call audit logging.
- Implemented a database-configurable multi-provider LLM router (Gemini, Ollama, OpenAI, Groq, Hugging Face) with per-policy retry and fallback logic.
- Designed a tool system (file I/O, PDF parsing, web search, object storage, code execution, transcript extraction) with per-agent tool allow-listing.

PolicyBot — Retrieval-Augmented Generation (RAG) Q&A Platform

A Q&A system that answers natural-language questions about internal documents — such as policies or leases — by retrieving the relevant passages and citing its sources.

[FastAPI](#) · [MongoDB](#) · [Qdrant](#) · [Gemini](#)

Independent Project · github.com/vishwajitvm/PolicyBot

- Built an end-to-end RAG query pipeline: vector and keyword retrieval, candidate merging, freshness-aware document resolution, citation generation, and composite confidence scoring.
- Implemented per-query execution tracing, persisting each pipeline stage to MongoDB for debugging and evaluation.
- Designed a pluggable vector-store interface (Qdrant implemented; Pinecone, Chroma, and MongoDB Atlas scaffolded) to support provider migration without touching business logic.

Tallyko — Multi-Tenant Restaurant / Retail POS Platform

A point-of-sale platform that lets multiple restaurants and retail stores independently manage billing, inventory, and kitchen orders from one shared system.

[FastAPI](#) · [PostgreSQL](#) · [Redis](#) · [React Native](#)

Independent Project · github.com/vishwajitvm/Tallyko

- Architected a multi-tenant FastAPI backend supporting both shared and dedicated-database tenancy models with Redis-cached tenant resolution.
- Built JWT authentication, Redis-backed sliding-window rate limiting, and a structured global exception-handling layer.
- Designed an offline-first mobile client (React Native + WatermelonDB) covering billing, inventory, kitchen display, and CRM workflows.

SmahrHR — AI Interview Platform

An AI system that conducts and scores first-round candidate interviews automatically, reducing manual recruiter screening workload.

[Python](#) · [LangChain](#) · [LangGraph](#) · [RAG](#) · [AI Agents](#)

Built at Bee Logical Software Solutions

- Architected an end-to-end AI interview pipeline using LangChain for LLM orchestration and LangGraph for stateful, multi-turn conversation management.
- Designed RAG pipelines that retrieve job descriptions and role requirements to generate targeted, role-specific interview questions.
- Built agent-based evaluation logic that scores candidate responses against defined criteria and generates structured interview summary reports.

OPEN SOURCE

TraceNest — Python Logging SDK

A logging library for Python/FastAPI applications that handles log rotation, retention, and automatic secret-masking, published for other developers to use.

Independent / Open Source · pypi.org/project/tracenest · github.com/vishwajitvm/TraceNest

- Designed and published a zero-dependency, thread-safe logging library with automatic day-wise rotation, retention enforcement, and background buffered flushing.
- Built a secret-redaction engine that recursively masks credentials, tokens, and database URLs across strings, dicts, and nested structures.

EDUCATION
